

## Lab Session: Building Python Applications with LLM APIs

**Duration: 4 Hours**

### Objectives:

- Understand the fundamentals of LLMs and their APIs.
  - Acquire and configure free API keys for LLM services.
  - Develop a basic Python application that interacts with an LLM API. [LinkedIn](#)
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### Session Breakdown:

#### 1. Introduction to LLMs and APIs (30 minutes)

- **Topics Covered:**
  - Overview of Large Language Models (e.g., GPT-4, Claude, LLaMA).
  - Differences between open-source and proprietary models.
  - Understanding API endpoints, authentication, and rate limits.

#### 2. Acquiring Free API Keys (40 minutes)

- **Activities:**
  - Guide students through the process of obtaining API keys from various providers:
    - **OpenAI free tier:** Offers a generous free tier to access GPT models such as 3.5 Turbo, 4o (Omni), 4o mini and whisper
    - **Google Gemini API:** Offers a generous free tier with access to the Gemini 1.0 Pro model.
    - **OpenRouter:** Provides a unified interface to access multiple free models, including DeepSeek R1.
    - **Eden AI:** Offers \$10 in free credits and access to various models like Claude 2 and LLaMA 2.
  - Discuss best practices for managing and securing API keys.

#### 3. Setting Up the Python Development Environment (30 minutes)

- **Activities:**
  - Install necessary Python packages (requests, dotenv, etc.).
  - Configure environment variables to securely store API keys.
  - Test API connectivity with a simple request.

#### 4. Developing a Basic LLM-Powered Application (80 minutes)

- **Project Options:**
  - **Chatbot Interface:** Create a simple chatbot that responds to user input using an LLM API.
  - **Text Summarizer:** Develop a tool that summarizes long articles or documents.
  - **Code Generator:** Build an application that generates code snippets based on user prompts.

#### 5. Testing and Optimization (40 minutes)

- **Activities:**
  - Test applications with various inputs to evaluate performance.
  - Optimize prompt engineering for better results.
  - Discuss handling API rate limits and errors gracefully.

#### 6. Wrap-Up and Q&A (20 minutes)